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Endophytic colonization and field responses of hybrid spruce seedlings after inoculation with plant growth-promoting rhizobacteria

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Abstract

Bacterial colonization and growth responses of hybrid spruce (*Picea glauca* × *engelmannii*) seedlings after inoculation with plant growth-promoting rhizobacteria (PGPR) were evaluated in controlled environment and field assays. Six antibiotic-resistant bacterial strains belonging to the genera *Bacillus* and *Pseudomonas* were used as inocula. For controlled environment assays, surface sterilized seeds were inoculated with log 6 *Bacillus* or log 8 *Pseudomonas* colony-forming units (cfu) under gnotobiotic conditions for assessment of rhizosphere and internal root and stem tissue colonization. In the field trial, one-year-old, container-grown hybrid spruce seedlings were inoculated with each of the *Bacillus* or *Pseudomonas* strains 1–2 days before planting at nine field sites in British Columbia and Alberta, Canada. All six strains colonized the rhizosphere with log 5.6−7.6 cfu g^{-1} root tissue under controlled environmental conditions. Two strains, *Bacillus* Pw-2R and *Pseudomonas* Sm3-RN, were also recovered from internal root and stem tissues with population sizes ranging from log 3.9 to log 5.0 cfu g^{-1} plant tissue. In the field trial, PGPR survived the winter in the rhizosphere with populations of log 2 to log 5 cfu g^{-1} root tissue 17 months after inoculation. In addition, spruce growth was significantly enhanced by bacterial inoculation at certain sites. The most effective growth-promoting strains were the endophytes, Sm3-RN and Pw2-R, and the external root colonizing strain, Ss2-RN. These three strains increased seedling dry weight up to 57% above noninoculated controls at five of the nine outplanting sites. *Pseudomonas* strain Sw1 and the other two *Bacillus* strains produced mean spruce dry weight increases at four of the sites. Seedling growth inhibition due to bacterial inoculation was detected at some sites. Our results confirm the short-term, site specific effectiveness of PGPR for reforestation of conifer seedlings, and indicate that benefits of a single inoculation at planting can extend through the second year in the field. In addition, bacteria capable of colonizing the seedling interior may be more effective PGPR for spruce than those restricted to the rhizosphere and the root surface. However, site specific seedling growth promotion may necessitate matching PGPR strains to outplanting sites for effective growth promotion.

Keywords: Spruce; Bacteria; Inoculation; Growth promotion; Endophytes

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1. Introduction

Forests are of global importance, providing numerous benefits such as fiber, pharmaceuticals, recreation and wildlife habitat. Global population pressures endanger the future of forests, as rates of harvesting commonly exceed those of regeneration. In addition, poor seedling survival and inadequate early growth performance of replanted sites further reduce the potential for successful forest regeneration. An inexpensive, environmentally benign alternative for enhancing productivity of newly established forest plantations involves nursery inoculation of seedlings with root-associated plant growth stimulating microorganisms. In forestry, this has traditionally been restricted to inoculation with mycorrhizal fungi to enhance seedling performance on sites where appropriate fungal symbionts are absent (Marx and Cordell, 1989; Kropp and Langlois, 1990). However, plant root systems, including mycorrhizae, are colonized heavily by asymbiotic soil microorganisms, of which bacteria are often most abundant (Rovira and Davey, 1974). Some of these root-associated bacteria have been shown to stimulate tree seedling growth and/or mycorrhizal infection in short term controlled environment and field experiments (Garbaye, 1994; Chanway, 1997), and may comprise an important component of the numerous soil biological factors that influence seedling growth. The term 'plant growth-promoting rhizobacteria' (PGPR) has been used to describe naturally occurring soil bacteria that stimulate plant growth after inoculation of seeds or roots (Kloepper and Schroth, 1978). Because bacterial inoculation of forest seedlings before outplanting would be an inexpensive and easily applied treatment, PGPR could be a useful, cost-effective tool for enhancing plantation establishment. In addition, some PGPR strains have recently been shown to be capable of colonizing internal plant tissues (Shishido et al., 1995; Hallmann et al., 1997). The comparatively intimate association of such endophytic bacteria with host plant cells may facilitate greater PGPR efficacy compared to rhizosphere associations, where microorganisms may be physically separated from root systems.

Here we report on colonization of external and internal hybrid spruce (*Picea glauca* × *engelmannii*) seedling tissues by PGPR, as well as survival and growth of this important British Columbia conifer species two growing seasons after inoculation and outplanting at nine field sites.

2. Materials and methods

2.1. Microorganisms

We used six bacterial strains previously shown to stimulate seedling growth: *Bacillus polymyxa* strains L6-16R, S20-R, and Pw-2R which are rifamycin-resistant derivatives of strains L6, S20, and Pw-2, respectively, and *Pseudomonas fluorescens* strains Sm3-RN, Ss2-RN, and Sw5-RN, which are nalidixic acid and rifamycin resistant strains derived from Sm3, Ss2, and Sw5, respectively (Shishido et al., 1996a, b). These antibiotic-resistant strains were similar to their respective wild-type strains in colony morphology, 23S-rDNA sequences, Biolog^{TM} (Biolog, Hayward, CA) carbon substrate utilization, GC-FAME profiles, and intrinsic antibiotic resistance to streptomycin, kanamycin, tetracycline, and nalidixic acid.

2.2. Seedling culture and bacterial inoculum for tube assays

Seedlings were generated from hybrid spruce seeds, supplied by Kalamalka Research Station and Seed Orchard, British Columbia Ministry of Forests (Vernon, B.C., Canada). Seeds were collected from a naturally generated hybrid spruce stand near Salmon Arm, B.C. (51°04′N, 119°26′), and stored at −17°C. No internal bacterial colonization was detected after surface sterilization, aseptic trituration and dilution plating.

For seed stratification, surface sterilized seeds were stored for 30 days at 4°C on moist, sterile gauze in sealed Petri plates. At the end of the stratification period, the effectiveness of this surface-sterilization procedure was confirmed by placing seeds on 10% strength tryptic soy agar (TSA) plates amended with 1% triphenyltetrazolium chloride for three days. Each germinating seed with no bacterial contamination was transferred aseptically to a glass tube (25 mm × 150 mm) containing moist (ca. 40% v v^{-1}), autoclaved montmorillonite clay aggregate 'Turface' (Applied Industrial, Deerfield, IL). Each seed was subsequently inoculated with 1.0 ml of a suspension containing one of the PGPR strains and 5 μg of rifamycin, and covered with Turface in the tube.

Bacterial inocula were prepared by growing *Bacillus* strains L6-16R, S20-R and Pw-2R in 50% strength tryptic soy broth (TSB) on a rotary shaker (120 rpm; 25°C) for three days. *Pseudomonas* strains Sm3-RN, Ss2-RN and Sw5-RN were grown under similar conditions, but in King's B broth instead of TSB. Bacteria were then harvested by centrifugation (ca. $3000 \times g$ for 10 min), washed and re-suspended in 10 mM sterile phosphate buffer, pH 7 (SPB) to a density of ca. log 6 cfu ml^{-1} for the *Bacillus* strains and log 8 cfu ml^{-1} for the *Pseudomonas* strains. Each bacterial treatment had 20 replicate tubes, at least eight of which yielded seedlings. The inoculation procedure was repeated two weeks later for all tubes containing seedlings.

All tubes were placed in a growth chamber (Conviron, CMP3244) under a 19 h photoperiod, photosynthetically active radiation at growth tube level of 160 μmol m^{-2} s^{-1}, and a 25 : 18°C light-dark temperature regime. Each tube was fertilized with 1.0 ml of a sterile solution containing (mg l^{-1}): N = 20.0, P = 5.0, K = 10.0, Ca = 10.0, Mg = 5.0, S = 9.6, Fe = 0.4, Mn = 0.05, Cu = 0.02, Zn = 0.02, B = 0.02, Mo = 0.005, and rifamycin = 5.0 every four weeks after sowing. Seedlings were grown five months before colonization of external and internal tissues was assessed.

2.3. Assessment of external and internal spruce colonization by PGPR

External and internal root colonization and internal stem colonization was evaluated on three, five-month old seedlings from each bacterial treatment. After gently removing seedlings from tubes, roots were separated from shoots and needles were removed from stems. To evaluate external root colonization, i.e., in the rhizosphere and on the rhizoplane, each root system was placed in a culture tube containing 10 ml of 10 mM SPB with glass beads and agitated for 1 min. Root washings were then serially diluted and plated onto 50% strength TSA amended with 200 mg l^{-1} rifamycin and 100 mg l^{-1} cycloheximide and 50 mg l^{-1} nystatin for *Bacillus*, and King's B agar (KBA) amended with 100 mg l^{-1} nalidixic acid, rifamycin and cycloheximide and 50 mg l^{-1} nystatin for *Pseudomonas*. Root washings were also plated on 50% strength TSA without antibiotics to check for contaminants in tubes.

To evaluate internal colonization by bacterial inocula, stems and root systems were washed individually and surface-sterilized by immersion in 1.8% NaClO for 1 min, followed by three rinses in sterile water. Tissues were then aseptically triturated in 10 ml SPB with a mortar and pestle, and serial dilutions of the resulting homogenates were plated (0.1 ml aliquots) on 50% strength TSA (for *Bacillus*) or KBA (for *Pseudomonas*) amended with antibiotics as described above. Dilution plates were incubated for up to four days at 28°C before colony numbers were assessed.

2.4. Field trial

One-year old, container-grown interior spruce seedlings were raised at forest nurseries in British Columbia and Alberta using standard operational protocol (Lavender et al., 1990). Outplanting trials were established during the spring of 1995 at nine sites representing a range of forest regions in which spruce occurs naturally (Table 1). Sites were 'paired' so that each region had a site with conditions anticipated to be more favorable for seedling growth than the alternate (Table 1). Seed provenances were carefully matched to outplanting sites to synchronize seedling phenology with regional daylength and climate. Seedlings of uniform height and appearance within each provenance were selected and inoculated on site 1–2 days before outplanting.

Seedlings were inoculated with the three *Pseudomonas* and three *Bacillus* strains described above. Inoculum was prepared using TSB cultures of each strain grown to stationary phase (1–2 days on a rotary shaker). Cultures were harvested by centrifugation ($3000 \times g$ for 15 min), washed in 20 mM sterile phosphate buffer (SPB) (pH 7.0), and resuspended in an equal volume of fresh SPB (ca. log 9 cfu ml^{-1}). Washed cultures were placed on ice and transported by air to each site for inoculation later in the day. Inoculum was diluted on site with water to approximately log 5 cfu ml^{-1} for *Bacillus* or log 8 cfu ml^{-1} for *Pseudomonas*. To inoculate seedlings, 5 ml of bacterial suspension were injected into the middle of the root plug using a sterile syringe and

Table 1
Location and anticipated relative productivity of outplanting sites used in this study

Site	Location	Biogeoclimatic zone (BGCZ)/Ecoregion (ER)[a]
TOT[b]	Cultivated field at the University of British Columbia, Vancouver, B.C.	Coastal Western Hemlock BGCZ (Vancouver Forest Region)
ASL[b]	Pine Ridge Forest Nursery, Smoky Lake, Alberta	Northern transition zone of Aspen Parkland ER (Aspen subregion) and the Boreal Mixedwood ER (Dry Mixedwood subregion)
ACL	Cold Lake, Alberta Reforestation Site	Boreal Mixedwood ER (Dry Mixedwood subregion)
WLR[b]	Williams Lake, B.C. Reforestation site	Interior Douglas-fir (IDF) BGCZ (Williams Lake Forest Region)
WLL	Williams Lake, B.C. Compacted Landing Site	IDF BGCZ (Williams Lake Forest Region)
PGF	Prince George, B.C. Fraser Road Site	Sub-Boreal Spruce (SBS) BGCZ (Prince George Forest Region)
PGS[b]	Prince George, B.C. Summit Lake Site	SBS BGCZ (Prince George Forest Region)
SMC[b]	Smithers, B.C. Blunt Creek Site	Engelmann Spruce–Subalpine Fir BGCZ (Smithers Forest Region)
SMS	Smithers, B.C. Shoe Horse Road Site	SBS BGCZ (Smithers Forest Region)

[a]According to Strong and Leggat (1981).
[b]Denotes sites anticipated to support higher seedling productivity.

needle. Control seedlings received 5 ml of diluted SPB.

Seedlings were outplanted within two days of inoculation using standard planting technique (Lavender et al., 1990), and were physically arranged according to a randomized complete block design. Seven treatments were evaluated at each site: uninoculated controls, seedlings inoculated with the three *Bacillus* strains and seedlings inoculated with three *Pseudomonas* strains. At each site, seven seedlings per treatment were planted in rows at 1 m spacing within each of 17–18 blocks. Seedling heights were recorded at the time of planting. In addition, a subsample of seedlings from each treatment was collected for preplanting root and shoot biomass measurements.

2.5. Data collection

Plantations were evaluated for survival and height growth of all seedlings in the fall of 1996, 17 months after inoculation and planting. In addition, one seedling per treatment per block was excavated at each site (i.e., $n = 17–18$) and transported to the University of British Columbia to assess biomass accumulation and PGPR root colonization. Entire seedlings with soil adhering to the root mass were stored frozen ($-20°C$) until processing. For biomass determinations, seedlings ($n = 10$) were separated into roots and shoots, and dried to a constant mass at 70°C. Root colonization by PGPR was evaluated on five spruce seedlings per treatment as described by Holl and Chanway (1992) and Shishido et al. (1996b). Briefly, root systems were pooled into groups of 2 or 3, and washed in SPB. Root washings were diluted serially, and aliquots of each dilution were plated in triplicate onto TSA agar amended with 200 mg l^{-1} rifamycin for *Bacillus* or 100 mg l^{-1} each of rifamycin and nalidixic acid for *Pseudomonas*. Selected seedlings were also assessed for internal root colonization using the surface-sterilization dilution plating assay described above.

2.6. Statistical analysis

A two-way ANOVA for a blocked design was performed on seedling growth data. Mean separation was accomplished using Fisher's protected least significant difference method. For root colonization by PGPR, the number of colonies that grew on antibiotic-amended TSA was related to the dry weight of root tissue and rhizosphere soil. These values were log transformed and subjected to ANOVA. The minimum detectable population size was ca. log 2 cfu g^{-1} soil. Root systems that produced no growth on dilution plates were assumed to be uncolonized and were assigned a value of zero.

3. Results

3.1. Recovery of PGPR from root and stem tissues in the controlled environment assay

Bacterial population sizes outside and inside roots and inside stems determined by dilution plating are presented in Table 2. Six of the 18 seedlings showed some fungal contamination in the rhizosphere, but no contaminants were found inside plant tissues. Strains L6-16R, S20-R, Ss2-RN and Sw5-RN were recovered from internal root tissues of only one (of three) seedlings assayed per treatment, and from a statistical perspective, did not develop population sizes different from zero. Though it cannot be concluded with certainty, we suspect that incomplete surface sterilization of root segments was responsible for these results. In contrast, internal root colonization was detected in all three seedlings treated with strains Pw2-R or Sm3-RN. These were also the only two strains that were recovered from surface-sterilized stem tissues in this assay.

3.2. Field trial

A range of seedling growth responses was detected 17 months after bacterial inoculation and outplanting, from highly positive (e.g. 82% increase in seedling biomass due to treatment with strain S20-R at site SMC) to negative (e.g. 31% decrease in biomass in response to strain Sw1-RN at site WLL) (Fig. 1). Overall, the best performing spruce inoculants in the field were the endophytes (strains Sm3-RN and Pw2-R) and the external root colonizing strain Ss2-RN. Each of these strains increased spruce biomass by up to 57% at five of the nine test sites. Due to the large degree of 'within treatment' variability, spruce biomass increases <28% were not statistically significant ($p < 0.1$). Seedling height growth followed trends similar to those observed for biomass (data not shown). Seedling survival was not affected by PGPR inoculation except at site SMS, where survival of inoculated seedlings was 21–36% greater than controls (data not shown).

PGPR were detected in the rhizosphere of spruce seedlings at most sites, sometimes with large (i.e., >log 6 cfu g^{-1} root tissue and rhizosphere soil) populations. Internal root colonization could not be confirmed due to incomplete root surface sterilization, but putative Pw2-R root endophytic populations of up to log 2.4 cfu g^{-1} root tissue were detected on seedlings from the PGS site (data not shown).

4. Discussion

We have previously reported pine and spruce seedling growth promotion in response to PGPR inoculation one year after bacterial treatment and outplanting (Chanway and Holl, 1993, 1994a, b). In the present study, we have determined that seedling growth promotion can extend through the second growing season, but with region- and site-specific influences. Statistically significant increases in spruce seedling growth were difficult to detect due to 'within treatment' variability, however, some large, significant biomass

Table 2
Recovery of PGPR from the rhizosphere and internal root and stem tissues of spruce five months after seedlings were inoculated

Inoculum	Rhizosphere (log cfu g^{-1} root)	Internal tissue (log cfu g^{-1} tissue)	
		Root	Stem
L6-16R	5.6 ± 0.3	0^{a}	0
Pw-2R	5.8 ± 0.5	4.4 ± 2.2	5.0 ± 0.2
S20-R	5.6 ± 0.4	0^{a}	0
Sm3-RN	7.6 ± 0.1	3.9 ± 1.9	4.6 ± 0.3
Ss2-RN	7.5 ± 0.1	0^{a}	0
Sw5-RN	7.1 ± 0.2	0^{a}	0

[a]One of three seedlings tested positive rendering the mean not significantly different from zero.

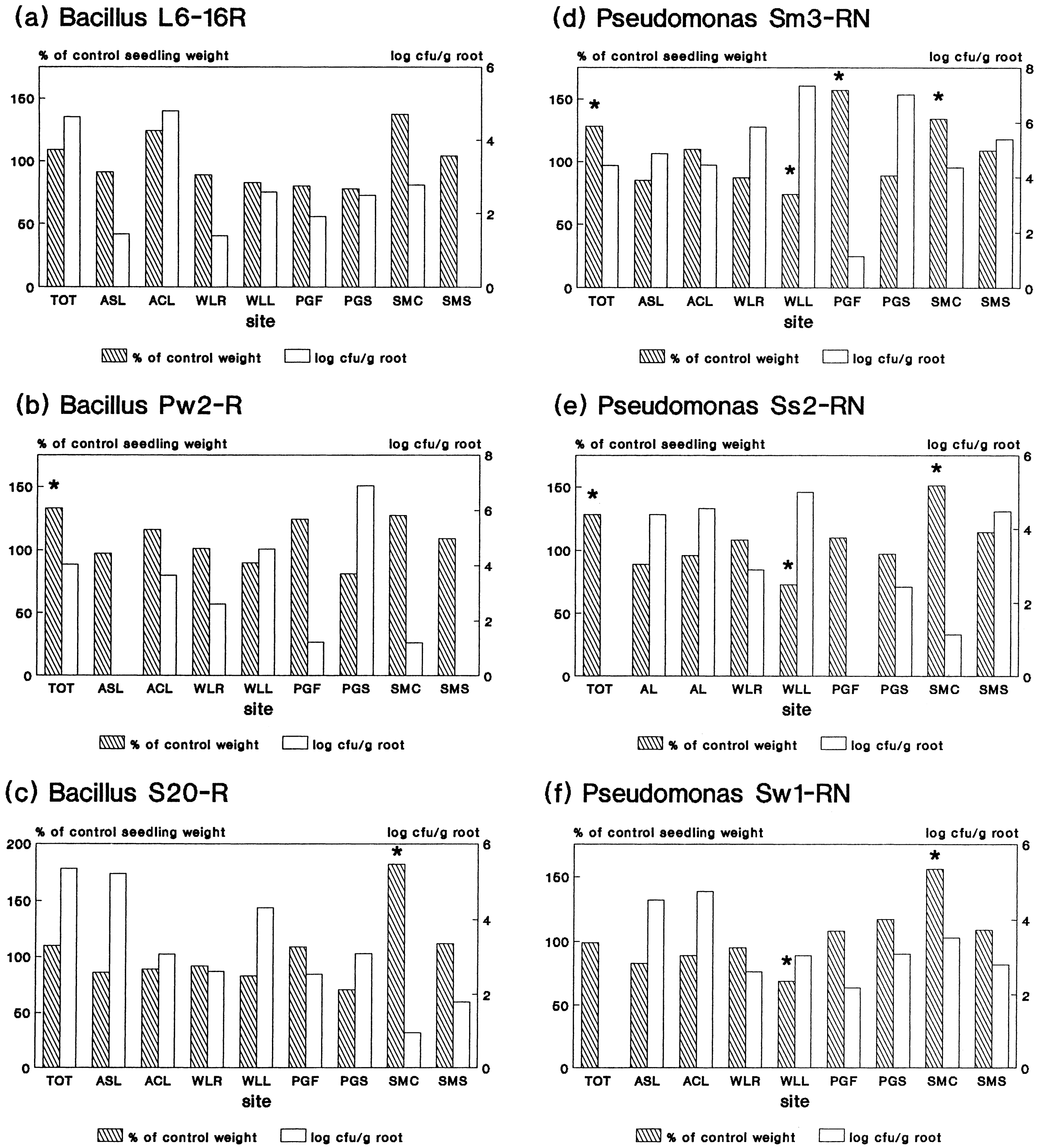

Fig. 1. Relative performance of PGPR-inoculated spruce seedlings, expressed as percentage of control seedling biomass, and root colonization by PGPR (log cfu g^{-1} root tissue and rhizosphere soil) 17 months after planting at nine field sites. (See Table 1 for site locations.) (a) *Bacillus* strain L6-16R, (b) *Bacillus* strain Pw2-R, (c) *Bacillus* strain S20-R, (d) *Pseudomonas* strain Sm3-RN, (e) *Pseudomonas* strain Ss2-RN and (f) *Pseudomonas* strain Sw1-RN. * Indicates statistically significant difference from control ($p < 0.1$).

gains were observed (e.g., Fig. 1 — site SMC). The forest region in which tests were conducted appeared to influence seedling growth responses to PGPR. For example, spruce seedling growth was enhanced to some degree by all PGPR at both Smithers sites, but inhibited by most PGPR at the Williams Lake sites (Fig. 1). Mixed results from other regions, with both positive and negative responses to inoculation, provided no evidence that PGPR efficacy was related to site quality. An analysis of site characteristics that may contribute to these differential responses is needed.

Most strains were recovered from spruce rhizosphere samples collected from these sites. However, some populations were quite low, i.e., ca. log 2 g^{-1} root tissue for strains Pw2-R and Ss2-RN at the SMC site. No general relationship between the size of root-associated PGPR populations and spruce growth responses was detected, but it is interesting to note that seedlings with the greatest biomass increases after inoculation (i.e., >50% above controls) supported the smallest numbers of PGPR in their root systems in three of four cases (see Fig. 1 — strains S20-R and Ss2-RN at the SMC site and strain Sm3-RN at PGF *versus* strain Sw1-RN at SMC).

It is also interesting that two of the three top performing PGPR strains are known to be able to colonize spruce seedlings systemically (Table 2) (Shishido et al., 1999). While internal shoot colonization was not evaluated in the field study, it is possible that the endophytic potential of these strains contributed to their plant growth-promotion efficacy. Further studies will elucidate the mechanism by which these strains enhance spruce growth, and the possible advantages of endophytic PGPR.

5. Conclusions

The results presented in this paper are part of an ongoing study of conifer seedling responses to PGPR inoculation, but some tentative conclusions can be drawn from our data. First, *Bacillus* and *Pseudomonas* PGPR can survive in the rhizosphere of outplanted spruce seedlings for at least 17 months. In some cases, inoculation can produce large, statistically significant increases in biomass accumulation after the second growing season in the field. PGPR efficacy may be facilitated by the ability of bacteria to colonize internal seedling tissues. However, 'within treatment' seedling growth response variability was a problem both within and between sites, and did not appear to be related to site quality. Finally, regional effects contributed, in part, to differential conifer growth responses to PGPR; seedlings planted in the Smithers forest region responded positively to PGPR inoculation, while those planted in the Williams Lake region did not. These findings suggest that some degree of PGPR strain $\times$ outplanting site matching may be necessary to use inoculation effectively on an operational basis.

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